

Raghav Senthil Kumar

623-332-2168 | raghavs9@illinois.edu | linkedin.com/in/raghav | github.com/raghav

EDUCATION

University of Illinois Urbana-Champaign

Champaign, IL

B.S. in Computer Science and Statistics

Expected: May 2029

- **Coursework:** Computer Science I (Java), Computer Science II (C++), Data Science & ML (Python), Calculus III
- **Activities:** Disruption Lab, SIGRobotics, AI Alignment, BuildIllinois, Innovation LLC, Agentic AI

EXPERIENCE

Founding Software Engineer | Clusion

September 2025 – January 2026

- Collaborated in an *Agile* team of **10 engineers** to design, develop and test the *Python (FastAPI)* search engine behind a SaaS research tool for high school debaters, deployed on the *AWS* cloud with **10K+ site visitors**
- Developed a retrieval algorithm that sifts a large data set of **600K+ documents**, filtering outliers with **IQR**
- Profiled, analyzed and optimized search on *Elasticsearch* so each query returns results in **under 1 second**
- Enhanced existing software with de-duplication logic that caught **30%** of stored documents as exact duplicates

Product Development Intern | Jetson

January 2025 – February 2025

- Tested 3 experimental features over **7 rounds of UI/UX testing** for **100K+ users**, proposing enhancements
- Redesigned onboarding in *Figma* from **12 pages to 5**, validating the enhancement with **A/B testing** on 30 users
- Suggested solutions to optimize app user engagement in a **5-page report** delivered directly to Jetson's CEO

Data Analytics Intern | STEM E

June 2024 – July 2024

- Used data science tools (*Python, pandas, Power BI, Excel*) to analyze **200+ TikTok posts** spanning 3 years
- Discovered that “science fun facts” posts earned 69.3% of all views despite being only 7% of what was posted
- Proposed a data-driven strategy that leadership adopted, raising overall engagement by **24% (8K+ views)**

PROJECTS

Chronos | [Live Website](#) | [Source Code](#)

December 2025

- Built a full-stack *JavaScript* web application (*React, Firebase*) that books meetings for groups of **30+ people**
- Wrote a **5-step Python** scheduling algorithm that scans availability, ranks open slots and emails confirmations
- Employed best practices for design and unit testing, writing a test plan of **14 unit and integration tests**
- Secured the app with a lobby that authenticates invited participants before queueing their follow-up meetings
- Delivered the software to **8 startup clients** with **2 LOIs (\$3.5K)**. **Won 2025 Congressional App Challenge**

SchoolPulse AI | [Live Website](#) | [Source Code](#)

June 2026

- Integrated and developed new AI capabilities that sift large data sets streaming from 3 edge AI hardware tools, running **4 Random Forest models** to detect anomalies in a school's live energy, water and waste usage data
- Engineered the *Python FastAPI* back end and application interfaces for a *Next.js* dashboard and *Gemma LLM*
- Led 5 teammates from their first idea through build and final pitch. **USAII Global Hackathon Finalist**

EcoBin | [Research Paper](#) | [Source Code](#)

March 2026 – May 2026

- Built a hardware/software smart bin prototype on an *Arduino Uno* that sorts each item in **under 2 seconds**
- Trained a two-stage *EfficientNetV2-S* network in *PyTorch* on **15K+ images**, reaching **96.13%** sorting accuracy
- Generated **9K** synthetic images with *U2-Net* and alpha-compositing. **2nd at Arizona State Science Fair**

Citation Chain | [Live Website](#) | [Source Code](#)

June 2026

- Developed an interactive *React Flow* web interface showing where a paper's citations land in **26 research fields**
- Implemented custom adapters for 2 application interfaces (*OpenAlex, Semantic Scholar*) to ingest paper data
- Wrote *SQL* queries over a *PostgreSQL* database to filter stored papers by research field, year and citation count
- Integrated a *Claude Sonnet 4.5 RAG* agent that sifts session-scoped vector documents to answer user questions

SKILLS

Programming Languages: Java, JavaScript, HTML/CSS, Python, SQL, C++

Frameworks/Platforms: React, Next.js, Node.js, FastAPI, Flask, Firebase, PostgreSQL, AWS, Elasticsearch, Vercel

DevTools: VS Code, Arduino IDE, Git, GitHub, Docker, CI/CD, Claude Code, Codex, Figma, Bun, Kaggle

Concepts/Practices: Unit Testing, Object-Oriented Programming, Web Development, Backend Development

Data & Analytics: pandas, NumPy, Power BI, Microsoft Excel, A/B Testing, Hootsuite